Milad Mohammadrezaei

Battery Diagnostics and Power Electronics Researcher

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Full name: Milad Mohammadrezaeikalehmasihi

Research Profile

Power electronics researcher with a Ph.D. and experience in lithium-ion battery health estimation, signal processing, and embedded systems. Research focuses on health indicators from partial charge/discharge data, transferable feature normalization, and online battery-parameter estimation. Head of R&D at Niak Energy Golsar, with hands-on development of DSP-controlled grid-connected inverters and a physics-informed machine-learning digital twin for fault prediction; inverter, EV-charger, and hybrid-storage products have reached TRL 8. Academic experience includes more than 800 hours of classroom teaching and B.Sc./M.Sc. mentoring. Recipient of the European Commission Seal of Excellence for the 2025 MSCA-PF proposal *iBattScan* (96/100); collaborating with Dr. Farshid Naseri on battery-health estimation.

Education

Isfahan University of Technology **Isfahan, Iran**
Ph.D. in Power Electronics, *GPA: 18.13/20; summa cum laude* *Sep 2018–Mar 2024*
Dissertation: *Physical Features Extraction for Online SoH Estimation of Lithium-Ion Batteries Based on Data-Driven Methods*
Supervisor: Prof. Ahmadreza Tabesh; **Co-supervisor:** Dr. Zeinab Maleki; **International advisor:** Dr. S. A. Khajehoddin (University of Alberta)

Isfahan University of Technology **Isfahan, Iran**
M.Sc. in Power Systems, *GPA: 17.50/20; summa cum laude* *Sep 2016–Aug 2018*
Thesis: *Techno-Economic Study of Collector Systems for Buoy Wave Energy Harvesters in the Persian Gulf*
Supervisor: Prof. Ahmadreza Tabesh; **Co-advisor:** Dr. A. Zamani

University of Isfahan **Isfahan, Iran**
B.Sc. in Electrical Engineering, *GPA: 15.36/20* *Sep 2012–Sep 2016*
Project: *Control Algorithms in Wind–Solar Hybrid Systems*
Supervisor: Dr. Hamidreza Kofigar

Research Interests

Battery Systems: Battery energy storage; SoH/SoC estimation; diagnostics from partial operating data; transferable health models; dynamic and relaxation features.

Control and Signals: System identification; advanced signal processing; battery-parameter estimation; physics-informed ML and digital twins for fault prediction.

Power Electronics: Grid-connected inverters; high-efficiency converters; DSP-based embedded control; hardware-in-the-loop prototyping.

Research Direction: Physical and electrochemical mechanisms of battery ageing; root-cause degradation diagnostics for next-generation battery management systems.

Lithium-Ion Battery Research

Health Estimation from Partial Operating Data: Developed physical health-feature extraction and selection methods for partial charge/discharge profiles, addressing battery operation without complete cycles. Evaluated the approach using the NASA Random Walk datasets with irregular operating profiles.

Transferable Online Health Estimation: Developed a framework that learns from other batteries to estimate the health of a battery without its own ageing dataset. Identified shared physical characteristics and normalized battery-specific differences to improve model transferability across cells and capacities. Evaluated the framework on NASA Random Walk, Oxford Battery, and CALCE CS2 and CX2 datasets; published in *IEEE Transactions on Transportation Electrification* (2024).

SoC–OCV Analysis and Parameter Estimation: Investigated changes in the state-of-charge–open-circuit-voltage (SoC–OCV) curve over battery lifetime. Developed an optimization algorithm for simultaneous estimation of internal parameters and state of health during ageing, and evaluated it using partial NASA battery data.

Dynamic and Relaxation Features: Investigated battery-health and lifetime estimation using relaxation and dynamic-response features. Collaborated with Dr. Farshid Naseri on fusion of ageing features from dynamic and rest conditions, resulting in an accepted IECON 2026 paper.

Chemistry-Related Features: Completed a simulation study using chemistry-related parameters independent of SoC in an ML algorithm for battery lifetime and parameter estimation. Evaluated the method on NASA Random Walk data to reduce dependence on whether operating records contain partial or complete charge/discharge profiles.

Industrial Battery Monitoring: Developed MATLAB-based software for monitoring and analysing lead-acid and lithium-ion batteries, and battery-parameter estimation algorithms for an industrial company in Iran. Experience also includes battery-system design and thermal analysis for energy-storage applications.

Technical Competencies

Battery Diagnostics: SoH/SoC estimation; partial charge/discharge feature extraction; cross-cell normalization; SoC–OCV analysis; parameter optimization; dynamic and relaxation features; thermal analysis.

Inverter Control: Single-phase HERIC inverters; TI TMS320F28066; sinusoidal PWM (SPWM); phase-locked loops (PLL); maximum power point tracking (MPPT); Park transformation; grid power injection.

ML and Digital Twins: Physics-informed ML; digital-twin implementation; prediction of current/voltage imbalance, DC-link faults, and fan- and temperature-related faults.

Embedded Systems: DSP/FPGA-based systems; ARM/AVR/Arduino; embedded converter control; hardware/software validation.

Programming: MATLAB/Simulink, C/C++, Python, DSP programming, LabVIEW; familiarity with Verilog/FPGA and Assembly.

Simulation and Design: PSIM, COMSOL Multiphysics, ANSYS Maxwell, PVsyst, LTspice, PLECS, EPLAN.

Communication: Modbus RTU/TCP, RS485/RS232, CAN bus, I2C, SPI.

Research Practice: Scientific writing with LaTeX; experimental protocols; laboratory safety; student mentoring; interdisciplinary R&D coordination.

Publications and Research Outputs

Published Peer-Reviewed Journal Articles

2025: **M. Mohammadrezaei**, Z. Maleki, and A. Tabesh, *A Correlation-based Method for the Estimation of the State-of-Health of Lithium Batteries in Electric Vehicles Using Data-Driven Methods*, **Energy Engineering and Management**, vol. 14, no. 2, pp. 2–11, 2025.
Contribution: Electrochemically informed feature selection for battery state-of-health estimation.

2024: **M. Mohammadrezaei**, Z. Maleki, A. Tabesh, and S. A. Khajehoddin, *A Framework for Normalizing Physical Features of Li-Ion Batteries to Form a Generic Health Estimation Model*, **IEEE Transactions on Transportation Electrification**, vol. 10, no. 3, pp. 6880–6892, 2024. DOI: 10.1109/TTE.2023.3331529.
Contribution: Capacity- and size-independent physical features for transferring battery SoH models across cells.

Accepted Peer-Reviewed Conference Paper

2026: **M. Mohammadrezaei** and F. Naseri, *A Hybrid State-of-Health Estimation Technique for Lithium-Ion Batteries Based on Fusion of Aging Features from Dynamic and Rest Conditions*, **IECON 2026—52nd Annual Conference of the IEEE Industrial Electronics Society**, Doha, Qatar, 18–21 October 2026. [**Accepted for presentation**]
Contribution: Fusion of ageing features from dynamic and rest conditions for battery-health estimation.

Published Peer-Reviewed Conference Paper

2024: **M. Mohammadrezaei**, A. Tabesh, and Z. Maleki, *Correlation-Based Methods for Data-Driven Lithium Battery State-of-Health Estimation in Electric Vehicles*, **International Conference on Energy Engineering and Management**, 2024. SID record: 1539084.
Contribution: Correlation-based selection of interpretable battery-health indicators.

Manuscripts Under Review

Aug 2026: **M. Mohammadrezaei**, A. Tabesh, Z. Maleki, and S. A. Khajehoddin, *Online Simultaneous Estimation of SoH and Internal Parameters of Lithium-Ion Batteries Considering the SoC–OCV Curve*, **IEEE Transactions on Industrial Informatics**. [**Under review; submitted August 2026**]

Jul 2026: **M. Mohammadrezaei** and H. Farzanehfard, *Fully Soft-Switched High Step-Up/Down Bidirectional Buck–Boost Converter*, **IEEE Transactions on Power Electronics**. [**Under review; submitted July 2026**]

Research Proposals and Collaboration

2025 Call: **Horizon Europe MSCA Postdoctoral Fellowships:** *iBattScan*, proposal No. 101285933, with Aalborg University, Denmark. Awarded the **European Commission Seal of Excellence**, with an evaluation score of **96/100**, on the first MSCA-PF submission; not funded.

Proposed Diagnostics: Developed an *operando* electrochemical impedance spectroscopy (EIS) concept for battery-condition assessment in *iBattScan*, targeting integration with operating energy-storage systems. The concept remains at the proposal stage and has not yet been experimentally implemented.

2026 Proposal: Continued refinement of *iBattScan* for HORIZON-MSCA-2026-PF-01, with Dr. Farshid Naseri as proposed supervisor at Aalborg University. Remote research collaboration has resulted in the accepted IECON 2026 paper listed above.

Research and Industry Experience

Niak Energy Golsar **Isfahan, Iran**
Head of Research & Development (full-time) *Jan 2022–Present*

- Designed and built a **5 kW single-phase grid-connected HERIC inverter** using the **Texas Instruments TMS320F28066** microcontroller.
- Developed the HERIC inverter firmware using **sinusoidal pulse-width modulation (SPWM)**; implemented a phase-locked loop (PLL) for grid synchronization, maximum power point tracking (MPPT), the Park transformation, and control of power injection into the grid.
- Implemented a **physics-informed machine-learning digital twin** for fault prediction, covering current and voltage imbalances, DC-link faults, and fan- and temperature-related faults.
- Lead development and validation of grid-connected and off-grid inverters, including 25 kW three-phase T-type designs.
- Lead development of 7 kW AC and 100 kW DC EV chargers and hybrid energy-storage systems exceeding 5 kWh; products have reached TRL 8.
- Carry out system design, DSP-based control, battery thermal analysis, hardware/software validation, and development of ML-based battery-monitoring tools.

Isfahan University of Technology **Isfahan, Iran**
Senior Researcher (part-time) *Sep 2024–Sep 2025*
Research supervisor: Prof. H. Farzanehfard.

- Developed high-efficiency power-converter topologies, analytical loss models, and soft-switching design guidance for solar and EV applications.
- Mentored two M.Sc. and three B.Sc. projects on DC–DC converters, grid-connected inverters, and lithium-ion battery monitoring.

Golnoor **Isfahan, Iran**
R&D Engineer (part-time) *Nov 2021–Jan 2022*

- Designed and built a 1 kW off-grid inverter from prototype to pilot production.
- Contributed to portable PV systems, solar-plant layouts, and Modbus/RS485 monitoring.

Teaching and Student Mentoring

Teaching Record: **More than 800 hours** of direct classroom teaching across two universities, including preparation of course materials and assessments, grading, and tutorials.

University of Isfahan **Isfahan, Iran**
Lecturer and Laboratory Instructor (part-time) *Sep 2023–Sep 2025*
Lectured in Electrical Circuits Theory and Electronics; instructed the Electrical Machines Laboratory for approximately 20 students per semester.

Isfahan University of Technology **Isfahan, Iran**
Lecturer and Teaching Assistant (part-time) *Sep 2018–Mar 2024*
Lectured in Fundamentals of Electrical and Electronics Engineering and Electrical Circuits; served as teaching assistant for Electrical Machines I and II.

Isfahan University of Technology **Isfahan, Iran**
Electrical Machines Laboratory Instructor (part-time) *Sep 2018–Sep 2023*
Supervised approximately 50 students per semester, prepared laboratory manuals, and enforced standard test procedures and safety practices.

Ph.D. Mentoring
2018–2023: Mentored four M.Sc. projects on battery electrochemical impedance spectroscopy (EIS), state-of-charge estimation using an extended Kalman filter, and battery-parameter estimation for EV driving cycles; guided junior students in power-electronics laboratory work.

Honors and Awards

2025 Call: **European Commission Seal of Excellence**, MSCA-PF *iBattScan*, 96/100.

2018: Ranked **1st** among approximately 40,000 candidates in Iran's National Ph.D. Entrance Examination in Electrical Engineering; admitted directly to IUT's Ph.D. programme based on academic excellence; ranked **3rd** in the M.Sc. Power Systems specialization at IUT.

2016: Ranked **82nd** among approximately 50,000 candidates in Iran's National M.Sc. Entrance Examination.

2012: Placed in the **top 5%** of Iran's National University Entrance Examination, among more than 500,000 candidates.

Academic Service and Memberships

Peer Review
Nov 2024–Aug 2026: Completed **16 verified peer-review reports covering 10 manuscripts** for *IEEE Transactions on Transportation Electrification*, *IEEE Transactions on Industrial Electronics*, and *AUT Journal of Electrical Engineering*.
Web of Science ResearcherID: LUZ-7277-2024.

PESI: Member, Power Electronics Society of Iran, 23 December 2024–10 August 2025.

IEEE: Member, Institute of Electrical and Electronics Engineers, 11 February 2023–20 November 2024.

Languages and Certification

Languages: Persian (native); English.

27 Jul 2019: **MSRT English language certificate**, Ministry of Science, Research, and Technology.

Academic References

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